

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K₄ Fe (CN)₆. After filtering, the precipitate is dried and weighed. The reactions which occur are $\text{ZnS} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{S}$ $2\text{ZnCl}_2 + \text{K}_4\text{Fe}(\text{CN})_6 \rightarrow \text{Zn}_2\text{Fe}(\text{CN})_6 + 4\text{KCl}$ If a 2 gram sample of rock yields 0.969 gram of Zn₂Fe(CN)₆, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn₂ Fe (CN)₆ = 342.6.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

<tool_call> reasoning process here

We are given:

- A 2 gram sample of rock yields 0.969 grams of Zn₂Fe(CN)₆.
- We need to find the percentage of Zn in the original rock sample.
- The molecular weight of Zn₂Fe(CN)₆ is given as 342.6 g/mol.
- The atomic weight of Zn is 65.4 g/mol.

Step 1: Find how many moles of Zn₂Fe(CN)₆ are produced.

Moles of Zn₂Fe(CN)₆ = mass / molecular weight
= 0.969 g / 342.6 g/mol
= 0.002828 mol (approximately)

Step 2: Each mole of Zn₂Fe(CN)₆ contains 2 moles of Zn.

So, moles of Zn = 2 × 0.002828 = 0.005656 mol

Step 3: Mass of Zn = moles × atomic weight
= 0.005656 × 65.4
= 0.3702 g (approximately)

Step 4: Percentage of Zn in the sample = (mass of Zn / mass of sample) × 100
= (0.3702 / 2) × 100
= 18.51%

This is closest to 18%

<answer> 18 % Zn </answer>